

ANNA UNIVERSITY, CHENNAI 600 025

CENTRE FOR DISTANCE EDUCATION

Faculty of Information and Communication Engineering



AI Based Age Prediction and Age Progression System Using Facial Images

A Project Report submitted in partial fulfilment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

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PROJECT GUIDE

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AgeVision is a browser-based, AI-driven platform that brings automated age estimation and identity-preserving facial age progression together in one secure interface. The client uses Angular 19; the server uses Django 5.2 with DRF 3.16, protected by JWT, Bcrypt and Fernet. A hybrid store keeps AI results in MongoDB (six collections) and logins in SQLite.

- Prediction uses two models together: MiVOLO v2 (about 3.65 yr error) with an InsightFace backup (about 8.5 yr).
- This is about 48 to 54% more accurate than How-Old.net and Face++ (6 to 8 yr).
- Progression uses three engines: SAM GAN, Fast-AgingGAN and FADING diffusion.
- Every aged face passes a FaceNet identity check (score above 0.6), met by over 95% of samples.
- The platform has 13 modules, 21 REST endpoints and 7-class emotion reading.
- It works as a reusable base for forensic, healthcare, entertainment, security and research use.

3.65 yr
average age error

> 95%
keep identity

13
modules

21
REST endpoints

Background

- More fields now use computers to read faces, such as apps, hospitals and police work.
- Two needs keep coming up: guess a person's age from a photo, and show how the face may look at another age while keeping the same identity.
- Human guesses differ from person to person, paid cloud tools cost money and raise privacy worries, and most free tools need developer skills.
- Doing both jobs well, for free, in one website, was not available before.

Limits of existing tools

- **How-Old.net** and **DeepFace** only guess age; they cannot age a face.
- **Face++** is accurate but paid, and runs only in the cloud.
- **AgingBooth** can age a face but is mobile-only and does not check identity.
- **Open-source libraries** mostly do prediction only and need developer set-up.

The gap AgeVision fills

AgeVision is one free, self-hosted web app that both guesses age and ages a face, keeps every photo on your own computer, and adds group, batch, emotion, live camera, history, analytics and an admin panel in a single login.

Objectives

- 1 Build a web app (Angular 19 + Django 5.2) that guesses and ages faces from one photo, many photos, or a live camera.
- 2 Guess age using two models together: MiVOLO v2 (main, about 3.65 yr error) and InsightFace (backup, about 8.5 yr error).
- 3 Offer three ways to age a face: SAM GAN (main), Fast-AgingGAN (runs on CPU), FADING diffusion (ages and de-ages; optional cloud GPU).
- 4 Detect 7 emotions on every face along with the age.
- 5 Keep the app safe with JWT, Bcrypt and Fernet, and show live progress for slow jobs.
- 6 Store data in two databases working together: MongoDB and SQLite.
- 7 Add an admin-only page with charts, user control, and live health checks.

Scope

- Sign up, log in, reset password, edit profile.
- Guess age for one face, a group photo, or many photos at once.
- Detect emotion with every age guess.
- Age a face while keeping the same identity.
- Personal history, charts dashboard and settings.
- Admin page for users, charts and system health.
- API can be used by mobile apps or other systems.

Six topics shaped the design, and each lesson from past work led to a clear choice in AgeVision.

Guessing age with deep learning

A 2024 NIST study found that single models stop improving at 4 to 7 years of error. So AgeVision uses two models together, namely MiVOLO v2 (about 3.65 yr) with an InsightFace backup.

Ageing a face with AI

Past work shows keeping the person's identity is the hard part. AgeVision checks identity with FaceNet (score above 0.6) and offers three engines: SAM, Fast-AgingGAN and FADING.

Finding and recognising faces

YOLOv8 finds faces first; RetinaFace is the backup. ArcFace codes help check identity, and OpenCV prepares both uploaded photos and live video.

Reading emotions

A ViT model from HuggingFace sorts each face into 7 emotions. It loads once and re-uses the faces already found, so it adds no extra work.

Clear results and two-way ageing

Earlier studies asked for confidence scores and the ability to both age and de-age. AgeVision shows a confidence value and uses FADING for two-way ageing; visual heatmaps are planned next.

Web frameworks for AI apps

Research recommends Django REST for AI apps. AgeVision pairs it with Angular 19 to keep the AI separate from the screens. Rival tools each do only part of the job.

AgeVision brings everything into one free, self-hosted web app behind a single login.

What AgeVision offers

- High-accuracy age prediction from two models working together.
- Three engines to age or de-age a face, all keeping identity.
- Emotion reading on every face along with the age.
- Many photos at once (batch) and live camera input.
- Personal history, charts dashboard and settings.
- Admin panel with user management and live health checks.
- 21 documented REST endpoints for outside apps.

Main uses

- **Forensics:** age progression to help find missing people.
- **Healthcare:** compare biological and chronological age.
- **Security:** age-gated access checks.
- **Research:** bundled MiVOLO v2 and SAM GAN training steps.
- **Bulk work:** process large case files through the batch module.

Hardware (Table 2.1)

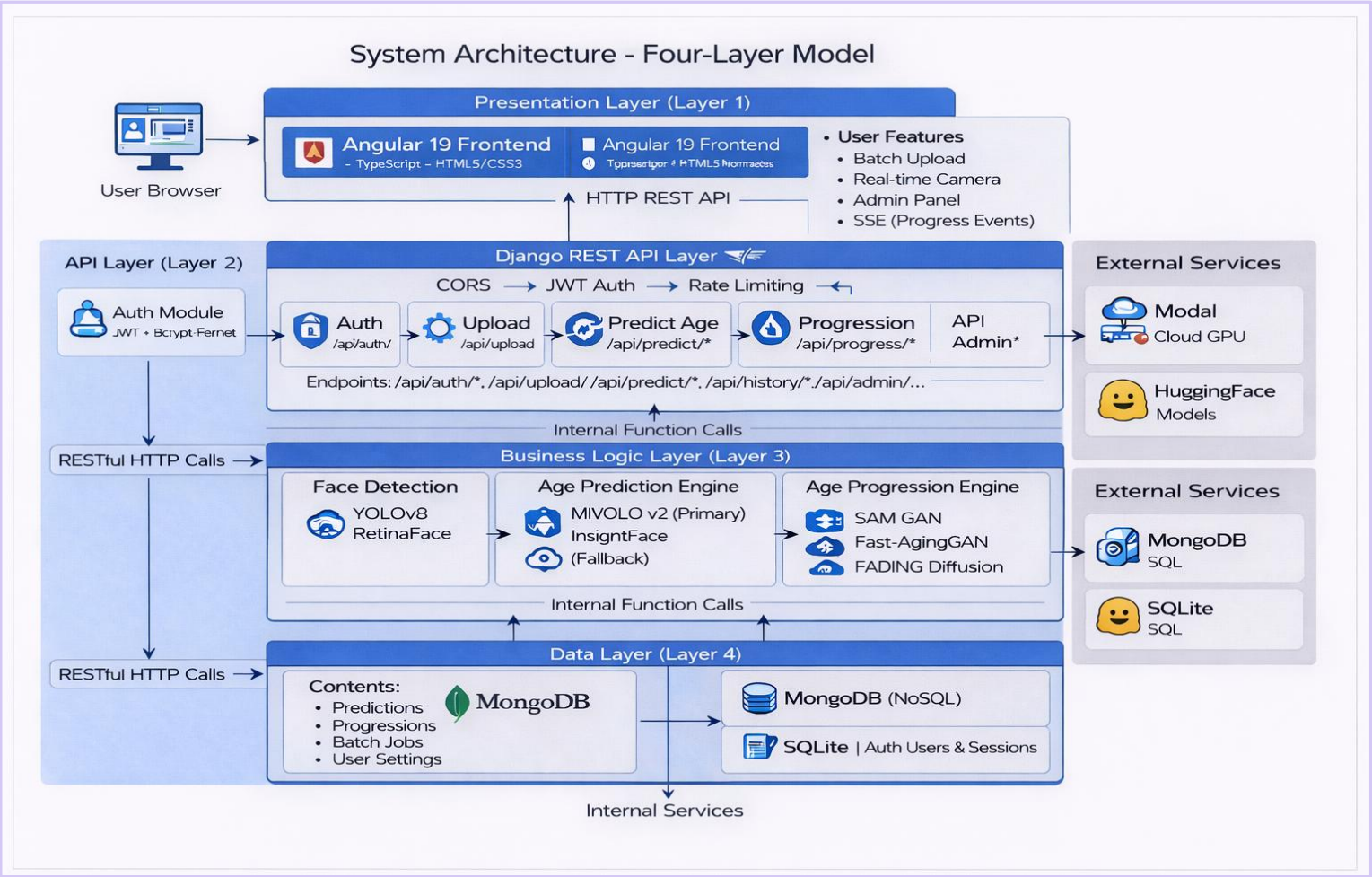
Component	Minimum	Recommended
Processor	i5 8th Gen / Ryzen 5	i7 / Ryzen 7 (8-core)
RAM	8 GB DDR4	16 GB+ DDR4
Storage	50 GB HDD	256 GB SSD
GPU	GTX 1060 (CUDA)	RTX 3070 / Tesla T4
Cloud GPU	None	Modal A10G / A100 (opt.)

Deployment notes

- Runs on Windows 10 or Ubuntu 20.04+ (Ubuntu 22.04 for live use).
- Needs internet to download model files (HuggingFace, Modal).
- FADING ageing works best with a CUDA GPU or a Modal cloud GPU.

Software (Table 2.2)

Part	Technology / Version
Frontend (screens)	Angular 19, Bootstrap 5.3, Chart.js 4, RxJS 7.8
Backend (API)	Python 3.11+, Django 5.2, DRF 3.16
Main age model	MiVOLO v2 (ViT + YOLOv8), ~3.65 MAE
Backup age model	InsightFace buffalo_1 (RetinaFace + ArcFace)
Ageing engines	SAM GAN, Fast-AgingGAN, FADING diffusion
Emotion / face finding	ViT face-expression, YOLOv8 + RetinaFace
Databases	MongoDB 4.6 (pymongo) + SQLite (Django ORM)
AI / security	PyTorch 2.0+, diffusers 0.25+, JWT+Bcrypt+Fernet



1. Screens (Presentation)

What users see, made with Angular 19, Bootstrap 5, Chart.js and RxJS.

2. API (Business logic)

Handles requests and security using Django 5.2 with DRF, JWT, CORS and 21 API calls.

3. AI / ML models

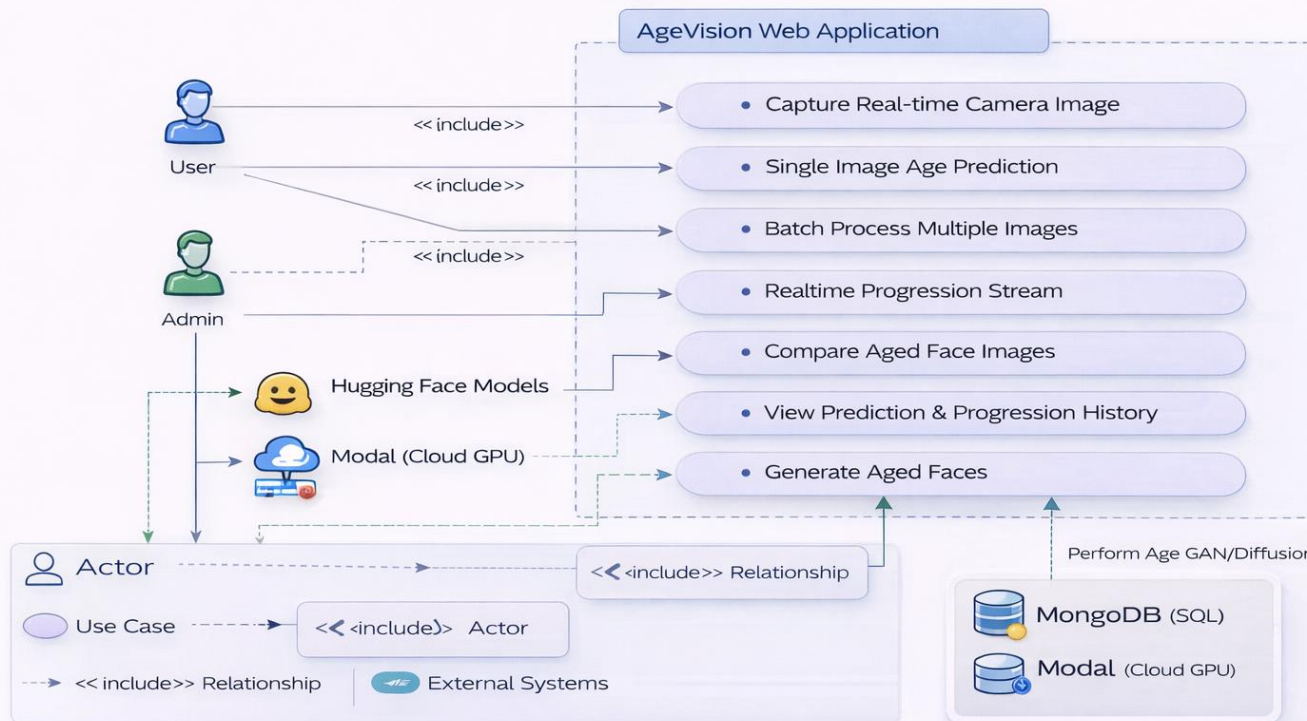
Runs the AI models, including MiVOLO v2, InsightFace, SAM GAN, Fast-AgingGAN, FADING, ViT emotion and YOLOv8.

4. Storage (Data)

Keeps data in MongoDB 4.6 (6 sets), SQLite (logins) and an optional Modal cloud GPU.

AgeVision - Use Case Diagram

AI-Based Age Prediction & Progression System



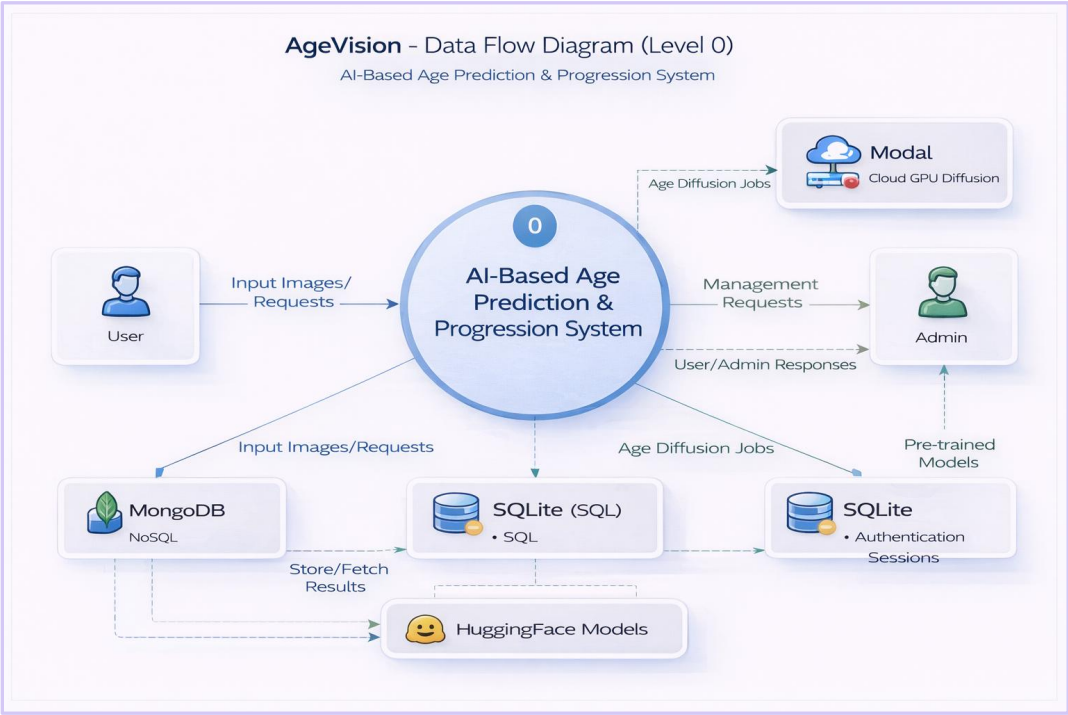
Four actors

- **User**: signs in and runs prediction, ageing, emotion and history.
- **Admin**: manages users and watches system health.
- **Platform**: the Angular client plus the Django backend.
- **External**: HuggingFace, YOLOv8 and Modal GPU.

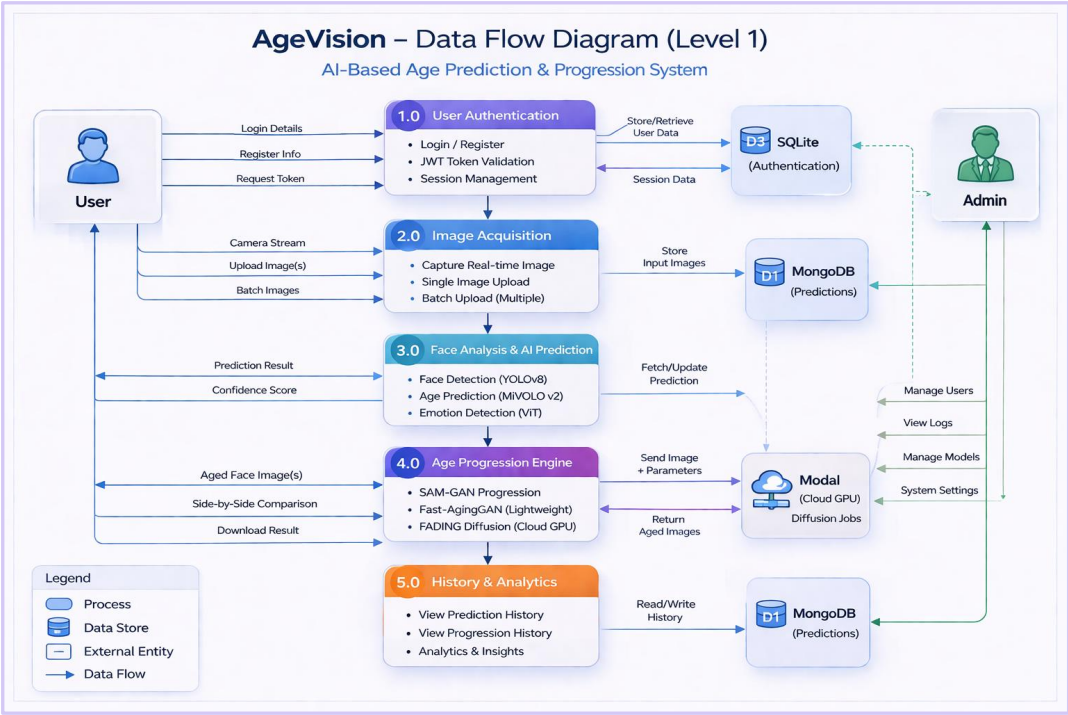
Main use cases

- Register, log in and reset password.
- Predict age for one, many or group photos.
- Age a face and read emotion.
- View history and charts.
- Admin: search, suspend and reinstate users.

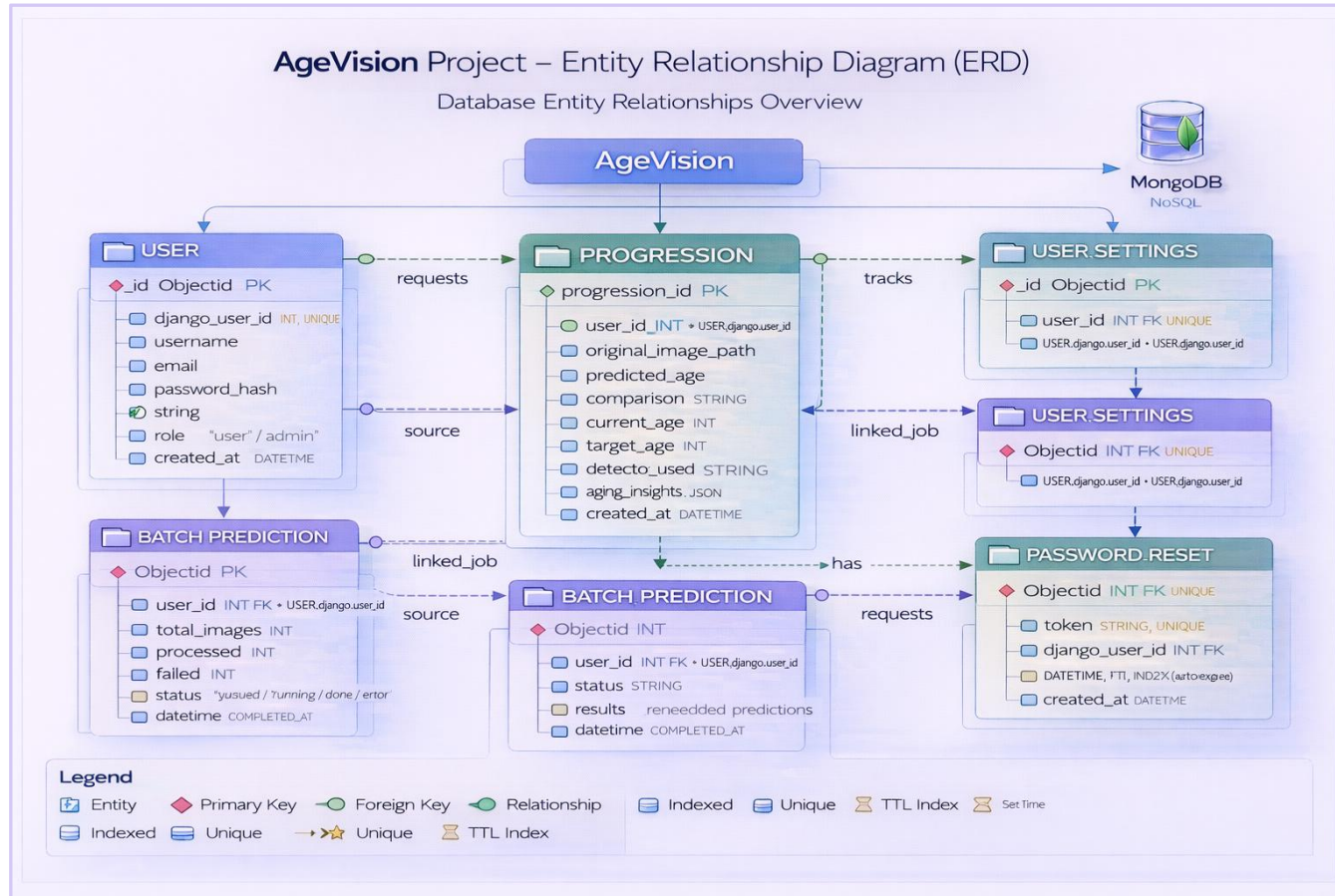
Level 0: the whole system



Level 1: eight sub-processes



Level 0 shows AgeVision as one process that takes photo uploads, batches, live camera frames and logins from the User and Admin, and returns ages, aged images and admin reports. Level 1 breaks it into eight steps: login, media upload, face finding, age and emotion, multi-engine ageing, history, analytics and admin. Each step owns its own MongoDB collection or SQLite table.



Hybrid storage

- **SQLite (Django ORM):** users, sessions and tokens.
- **MongoDB 4.6 (PyMongo):** all AI results in six collections.
- **One User** has many Predictions and Progressions, exactly one Settings, and many Batch Jobs.
- **Each Batch Job** groups many Predictions; each Prediction can hold many faces.

Six MongoDB collections

- predictions, progressions, user_settings,
- batch_predictions, password_resets, history.

Thirteen features, all built on Django and shown through the Angular 19 app.

1

Dashboard

Totals, recent results and quick actions.

3

Single-Face Age

Find one face and guess its age.

5

Many Photos (Batch)

Upload and process many images together.

7

Face Ageing

Age or de-age a face with three engines.

9

History

Saved predictions, progressions and batches.

11

Settings

Theme, preferred model and engine, alerts.

13

Admin Panel

User management and live system health.

2

Login & Sign-up

Register, log in and reset password (JWT).

4

Group Photo

Age every face in a group photo.

6

Emotion Reading

Label each face with 1 of 7 emotions.

8

Live Progress

Show a live progress bar for slow jobs (SSE).

10

Charts Dashboard

Counts, age spread, gender and emotion charts.

12

Live Camera

Read webcam frames and overlay age live.

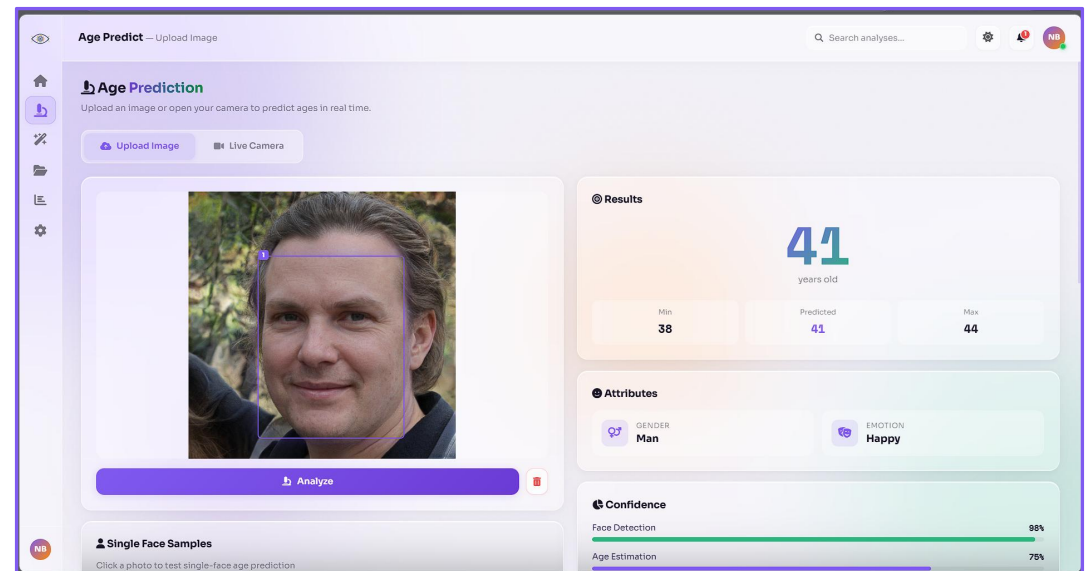
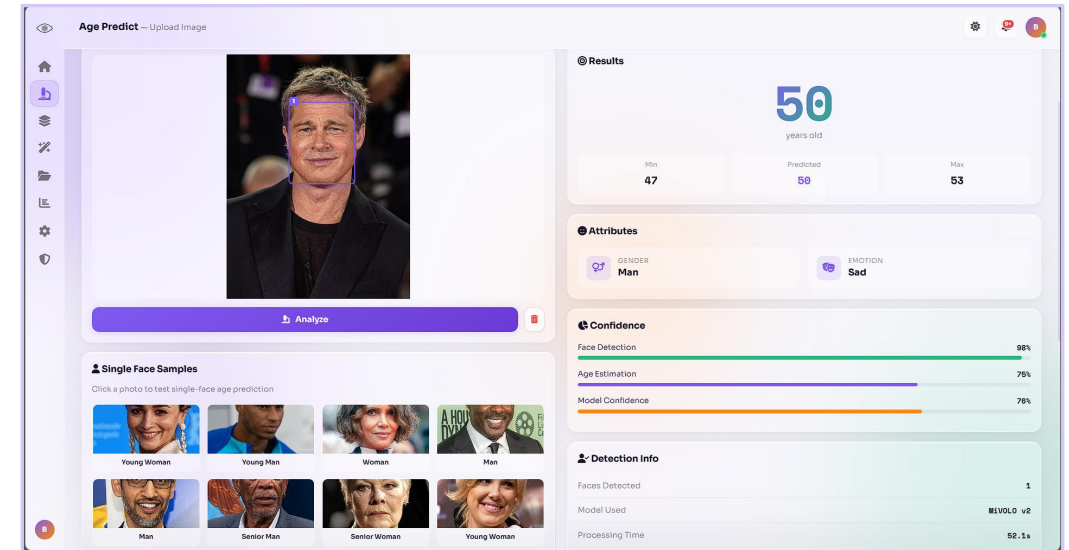
MiVOLO v2 + InsightFace

- **Main model: MiVOLO v2** looks at both the face and the body.
- **Backup model: InsightFace** is used only if needed (about 8.5 yr error).
- **Combined result** is 70% MiVOLO plus 30% InsightFace.
- **Finding faces** uses YOLOv8 first, then RetinaFace as backup.
- **Group photos** give the age of every face, with gender and emotion.
- **Confidence** shows how sure the result is, per face.
- **Live camera** reads a new frame every 1.5 seconds.

Average error about 3.65 years

About 48 to 54% fewer errors than How-Old.net or Face++ (6 to 8 yr)

Emotion: a ViT model sorts each face into 7 feelings (angry, disgust, fear, happy, neutral, sad, surprise), re-using the faces already found.



Pick one of three engines. Every result must pass a FaceNet identity check (score above 0.6).

SAM GAN

MAIN

Turns the photo into StyleGAN2 code, then shifts it toward the target age. Tuned for Indian faces. Takes 3 to 8 seconds.

Fast-AgingGAN

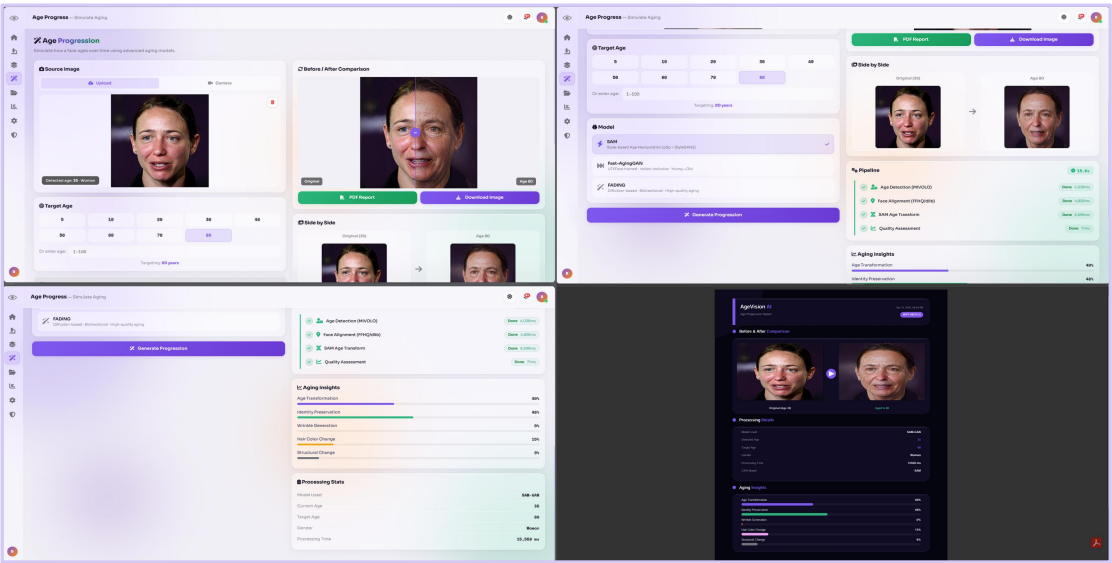
RUNS ON CPU

A small CycleGAN (about 11 MB). Works on a normal computer with no GPU, which is good for low-power setups.

FADING Diffusion

BEST QUALITY

Uses Stable Diffusion to age or de-age a face. Highest quality; can use a Modal cloud GPU.



> 95%

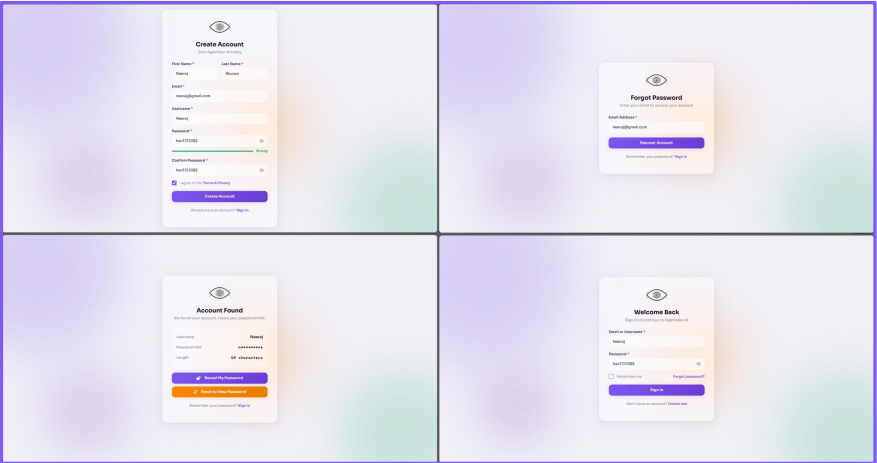
of faces keep the same identity

3 to 8 s

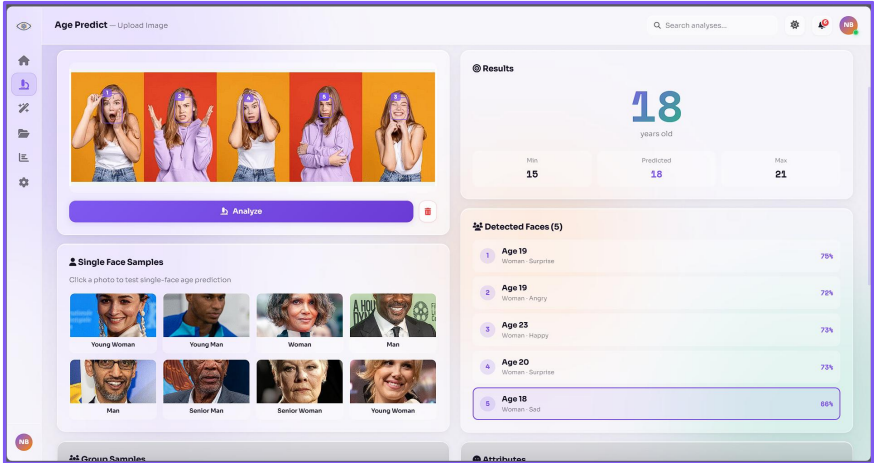
to age a face with SAM GAN

Two-way

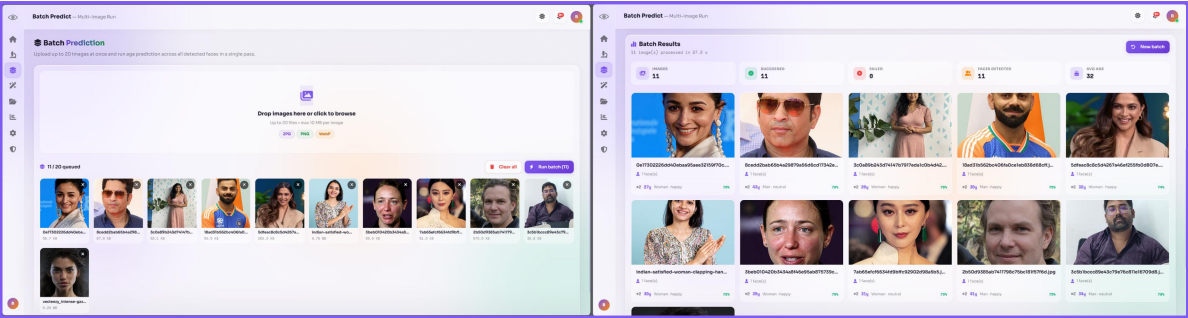
FADING can age and de-age a face



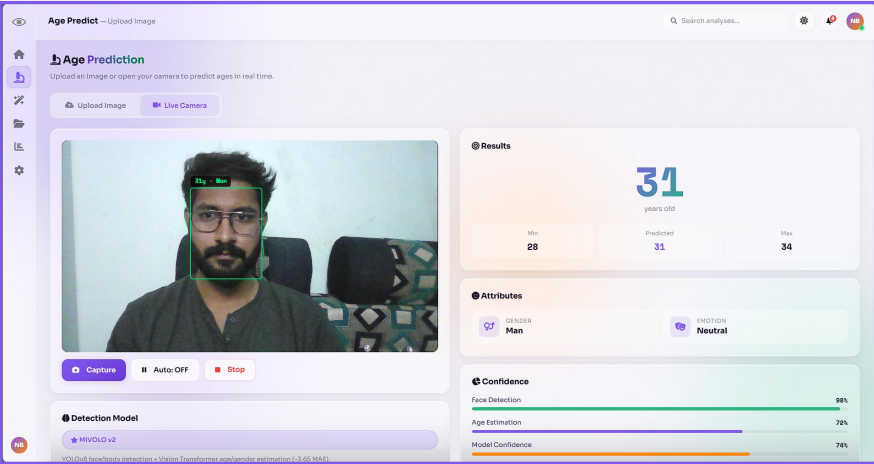
Secure login, sign-up and password reset



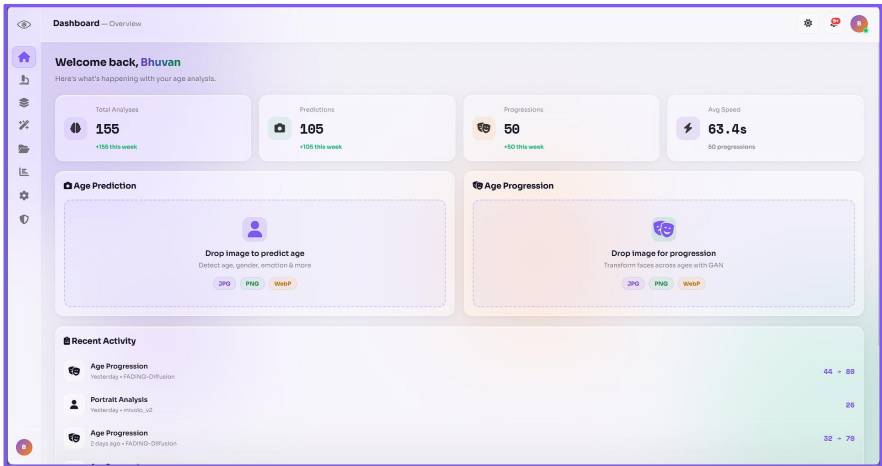
Group photo with age, gender and emotion



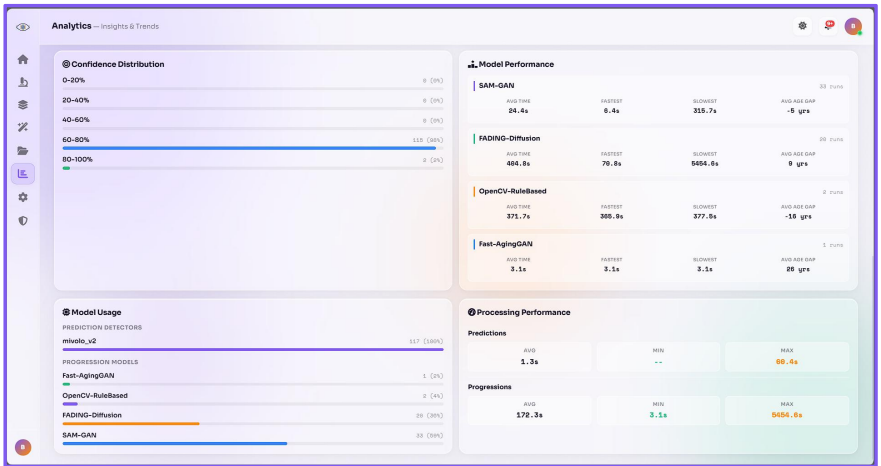
Batch: many photos processed together



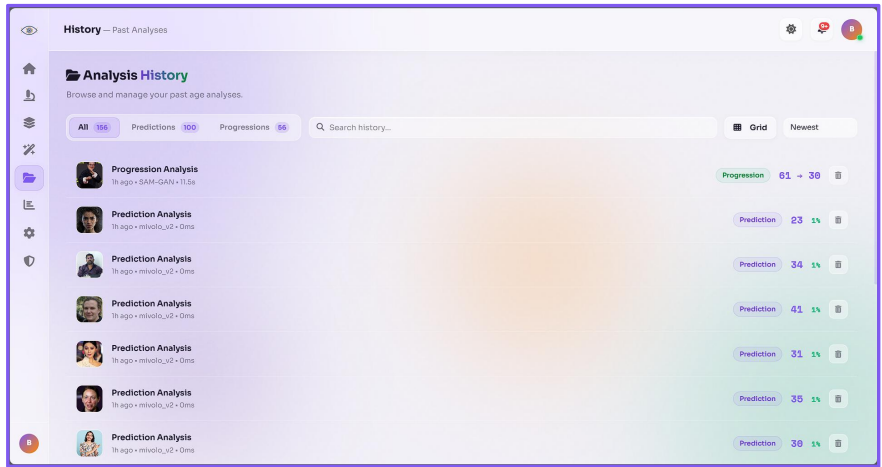
Live camera with age overlay



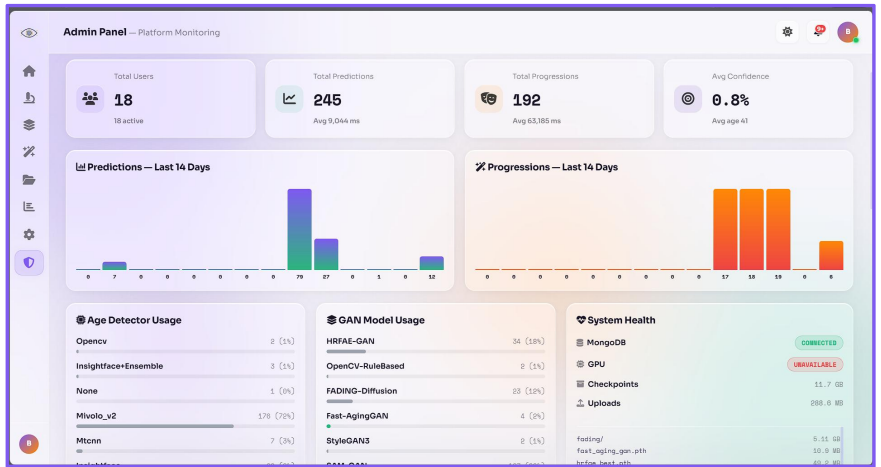
Home dashboard with totals and recent results



Charts for age, gender and emotion



History with thumbnails, download and delete



Admin: user management and system health

Datasets (Table 4.1)

Dataset	Size	Used For
UTKFace	23,000+	Train + test MiVOLO v2
IMDB-WIKI	500,000+	Extra training for MiVOLO v2
MORPH	55,000	Testing only
FFHQ	70,000	Train SAM GAN + FADING

An Indian-face subset (filtered with YOLOv8 confidence of at least 0.8) was added to FFHQ to improve results on South-Asian faces. MORPH is kept aside only for testing, never for training.

How the algorithms work

- **MiVOLO v2:** a two-head Vision Transformer that also reads body cues from YOLOv8. The final age is 70% MiVOLO plus 30% InsightFace.
- **SAM GAN:** a pSp encoder turns the photo into StyleGAN2 W+ code, then moves it toward the target-age direction.
- **FADING:** null-text inversion rebuilds the photo's diffusion path, then classifier-free guidance steers it to the target age, so it can age and de-age.
- **Identity check:** FaceNet cosine similarity between the old and new face must stay above 0.6.

What we checked in testing

Area	What it checks	Tests
Login (TC-A)	Sign-up, login, JWT, reset	10
Prediction (TC-P)	Single, group, camera	9
Batch (TC-P)	Many photos and saving	2
Ageing (TC-G)	SAM / Fast / FADING + progress	11
History (TC-H)	List, delete, user privacy	4
Charts (TC-N)	Totals and chart data	3
Settings (TC-S)	Theme, model, checks	4
Admin (TC-AD)	Stats, suspend, health	7

Speed (Table 4.5)

Step	Time grows with
Finding faces (YOLOv8)	O(n) by pixels
Guessing age (2 models)	O(f) by faces
SAM GAN ageing	O(1) per image
FADING ageing	O(T) by steps

50

tests, all passed

13

features built

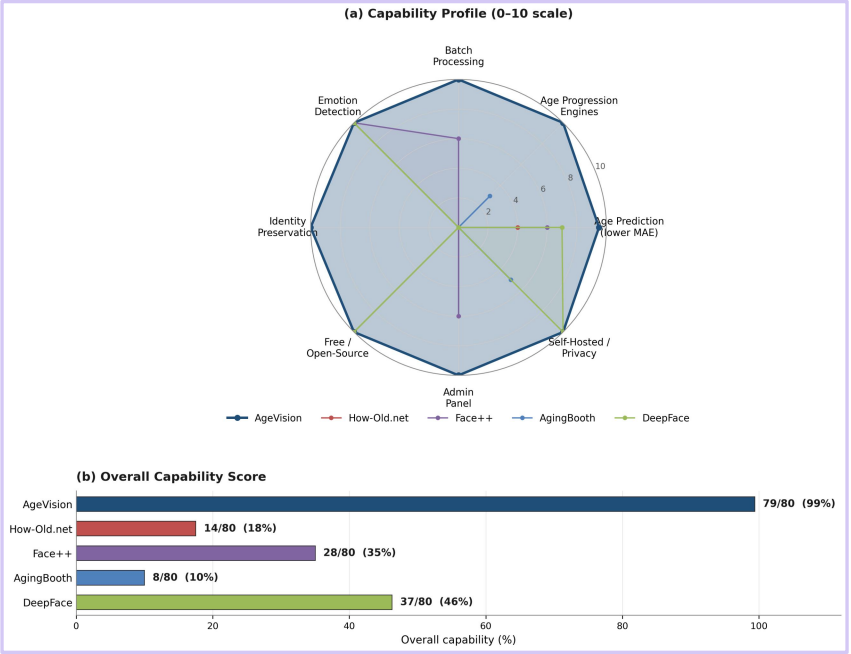
21

REST endpoints

Every module was checked against a written test catalogue of 50 cases covering normal use, wrong input and edge cases. All 50 passed.

Feature	AgeVision	How-Old	Face++	AgingBooth	DeepFace
Age error (MAE)	~3.65 yr	6 to 8 yr	5 to 7 yr	N/A	~5.4 yr
Ageing engines	3 (GAN+Diff)	None	None	1 (GAN)	None
Many photos at once	Yes	No	Paid	No	No
Emotion reading	7-class ViT	No	Yes	No	Yes
Keeps identity	FaceNet ≥ 0.6	No	No	No	No
Free / open-source	Yes	Yes	Paid	Paid	Yes
Admin + self-hosted	Yes	No	Paid	No	Partly

Table 4.6 shows that AgeVision is the only free, self-hosted tool that does all of this in one place.



Headline result

About 3.65 years of error, which is roughly 48 to 54% fewer mistakes than paid tools, and over 95% of aged faces keep the same identity.

What we achieved

- Two-model age guessing gives about 3.65 years of error (48 to 54% better than paid tools).
- Three ageing engines, with over 95% keeping the same identity.
- FADING can both age and de-age a face realistically.
- 13 features, 21 API calls and 6 MongoDB sets, all login-protected.
- Charts dashboard plus an admin-only control page.
- Re-usable training steps for MiVOLO v2 and SAM GAN included.

What's next

- Host fully in the cloud (AWS / GCP GPU) with a job queue for slow tasks.
- Add a PDF download for each result and a mobile app.
- Add visual heatmaps and Tamil / Hindi screens.
- Connect to police case systems and speed up admin charts.
- Auto-retrain the models as new photos arrive.
- Bundle model files for offline (no-internet) use.

Key references: MiVOLO (2025) • NIST FATE 8525 (2024) • Null-text Inversion, CVPR 2023 • SAM: Only a Matter of Style • Django REST Framework

Thank You and Open to Questions